Identifying NFL Quarterback Archetypes via K-Means Clustering

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Abstract

Quarterback classification remains central to NFL evaluation and team-building strategies, yet most archetype frameworks rely on subjective media narratives rather than quantitative analysis. This study applies unsupervised machine learning to identify quarterback playing styles using six seasons of performance data (2019-2024) from qualifying NFL quarterbacks. Principal Component Analysis combined with K-means clustering identified four distinct archetypes: Pocket Passer, Game Manager, Gunslinger, and Dual Threat, which align with established football schema while revealing significant boundary overlap between categories. One-way ANOVA tests confirmed statistically significant differences across all examined metrics (all $p < 0.001$), validating cluster meaningfulness despite a moderate silhouette score of 0.208. Analysis revealed that high-performing quarterbacks of different archetypes often display similar efficiency ratings while differing in other elements such as rushing involvement and pass volume. These findings suggest that while quarterback archetypes represent legitimate performance patterns, they exist along a spectrum or a continuum rather than as rigid discrete categories. This work provides a data-driven framework for player evaluation

and challenges the oversimplified archetype narratives prevalent in sports media and scouting.

Keywords: quarterback archetypes, K-means clustering, sports analytics, unsupervised learning, NFL performance metrics, player classification.

1 Introduction

The introduction needs to answer three questions to provide context and relevance:

1. Why should we care about this research (its importance)?
2. What has been done in the literature (identify a gap)?
3. What is new (your contributions)?

When citing literature, be sure to understand the difference of textual citations and parenthetical citations. Here are some examples.

Xie (2015) did something ...

Recently there has been an increase in the prevalence of diabetes in the United States (Wild et al., 2004). Notice that the citation is included *inside* the punctuation, as it is part of the sentence.

Considerable work has already been published in this area (e.g., Xie, 2015). Some parametric bootstrap sample size approach was proposed by Dwivedi et al. (2017).

Also notice the use of `\label-\ref` pairing within the source (.tex) document to enable cross-referencing of sections, tables, and figures.

The next paragraph is a roadmap for the rest of your manuscript. The roadmap paragraph should be the final paragraph in Section 1. Notice in the previous sentence that the 'S' in section is capitalized when we refer to a specific numbered section. Similarly the 'T' in table and 'F' in figure should be capitalized when referring to specific numbered tables and figures (see examples in Sections 2 and 4).

Table 1: This is my first table. Table captions should be located above the table.

	Var 1	Var 2	Var 3
Feature 1	−0.110	4	2.401
Feature 2	0.116	4	3.529
Feature 3	−0.066	6	11.104
Feature 4	0.219	3	4.815
Feature 5	0.303	5	2.188
Feature 6	0.544	0	8.050
Feature 7	−2.617	8	3.646

The rest of the paper is organized as follows. The data will be presented in Section 2. Section 3 describes the methods. The results are reported in Section 4. A discussion concludes in Section 5.

2 Data Description

Use this section to describe the data that helps to answer your research questions. Some descriptive statistics in tables or figures are suggested here to summarize data. See Notes 4 and 5 for details.

The response variable Y is the log-concentration of calcium in the blood, measured in mg/dL. Notice that we use $ to enclose display-math expressions, including just variable names.

Each table and figure included must be explicitly referenced and commented upon in the text. For example, Table 1 summarizes some distributional features for some of the variables in our dataset. The `[!t]` specification forces the table to be located at the top of the page.

3 Methods

Use this section to present the methodologies that will generate results by analyzing the data. Only the methods should be described in this section; the results go in the next section.

It is strongly advised **NOT** to combine Methods and Results into one section.

3.1 First Subheading

It may be wise to break up Section 3 into multiple subsections. Remember that good headings and subheadings are not only helpful when first organizing our thoughts and starting the first draft, but are also helpful in orienting the reader to the structure of the report. For example, feature selection, model selection and training, and evaluation metrics may be great subheadings.

This text is in blue. It is sometimes helpful to highlight (in color) the changes that were made in response to reviewer's comments.

3.2 Some Equation Examples

Recall Einstein's equation

$$E = mc^2, \tag{1}$$

which states that the energy E of a particle in its rest frame as the product of mass (m) with the speed of light squared (c^2). Equation (1) is a display-math expression.

Suppose that the radius of a circle is r. Then its area is given by

$$\pi r^2. \tag{2}$$

Equation (2) is interesting, although it is short. If the expression is not referenced elsewhere in the document, would likely be better as an inline expression, as in 'Then its area is given by πr^2.'

We can avoid numbering (labeling) a display-math expression by enclosing the expression with `\[` and `\]`, as in

$$f(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right),$$

which is the density of a standard normal variable. Equations and mathematical expressions that are not referenced elsewhere in the document should not be numbered (labeled).

The `align` environment allows for alignment of equations or expressions. We can also use `array` for matrices. As in tables, the & character is used for alignment. For example,

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu_+ \\ \sigma^\mu_- & 0 \end{pmatrix}, \qquad \gamma^5 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix},$$
$$\sigma^\mu_\pm = (\mathbf{1}, \pm\sigma),$$

giving

$$\hat{a} = \begin{pmatrix} 0 & (\hat{a})_+ \\ (\hat{a})_- & 0 \end{pmatrix}, \quad (\hat{a})_\pm = a_\mu \sigma^\mu_\pm. \tag{3}$$

We can use `align` for long expressions, too, as indicated by

$$\begin{aligned} \sum_{u \in C^+} \left\lfloor \frac{w'(u)}{m} \right\rfloor &\leq \left\lfloor \sum_{u \in C^+} \frac{w'(u)}{m} \right\rfloor \\ &\leq \left\lfloor \frac{r(v) + \sum_{u \in C^+} w(u)}{m} \right\rfloor = \left\lfloor \frac{w(v) - \sum_{u \in C^-} w(u)}{m} \right\rfloor \\ &\leq \left\lfloor \frac{w(v)}{m} \right\rfloor - \sum_{u \in C^-} \left\lfloor \frac{w(u)}{m} \right\rfloor = s(v) + \sum_{u \in C^+} \left\lfloor \frac{w(u)}{m} \right\rfloor \end{aligned} \tag{4}$$

and

$$\begin{aligned} \sum_{u \in C^-} \left\lfloor \frac{w(u)}{m} \right\rfloor &\leq \left\lfloor \sum_{u \in C^-} \frac{w(u)}{m} \right\rfloor \\ &\leq \left\lfloor \frac{r'(v) + \sum_{u \in C^-} w'(u)}{m} \right\rfloor = \left\lfloor \frac{w'(v) - \sum_{u \in C^+} w'(u)}{m} \right\rfloor \\ &\leq \left\lfloor \frac{w'(v)}{m} \right\rfloor - \sum_{u \in C^+} \left\lfloor \frac{w'(u)}{m} \right\rfloor = s'(v) + \sum_{u \in C^-} \left\lfloor \frac{w'(u)}{m} \right\rfloor . \end{aligned} \tag{5}$$

Notice that all mathematical expressions written above, both inline and display-math, are properly punctuated, as they are included as part of their respective sentences.

4 Results

Report the results generated from the methods discussed in the previous section. Results can be from simulation and/or real data analysis.

Focus on clearly presenting your findings from the data analysis, using a logical sequence, incorporating relevant tables and figures to effectively communicate key trends and patterns, while ensuring the information directly aligns with your research questions outlined in Section 1 and the methods described in Section 3.

Other results, tables and figures may be included in an appendix or supplemental file.

4.1 Results Subheading

It is likely a good idea to break your results into subsections (e.g., one for each evaluation metric.

4.2 Figures

Figure 1 shows the distance against the speed from this dataset. As in Table 1, the `[!t]` specification for Figure 1 forces the figure to be located at the top of the page.

5 Discussion

Answer the following questions in the Discussion (and Conclusion; see Note 5 for more details).

- What are the main contributions and key features or our analysis and results?
- How did this research address the 'need' identified in the introduction?
- Do our results (appear to) confirm or contradict our initial expectations?
- What are the generalizations and limitations of the current study?

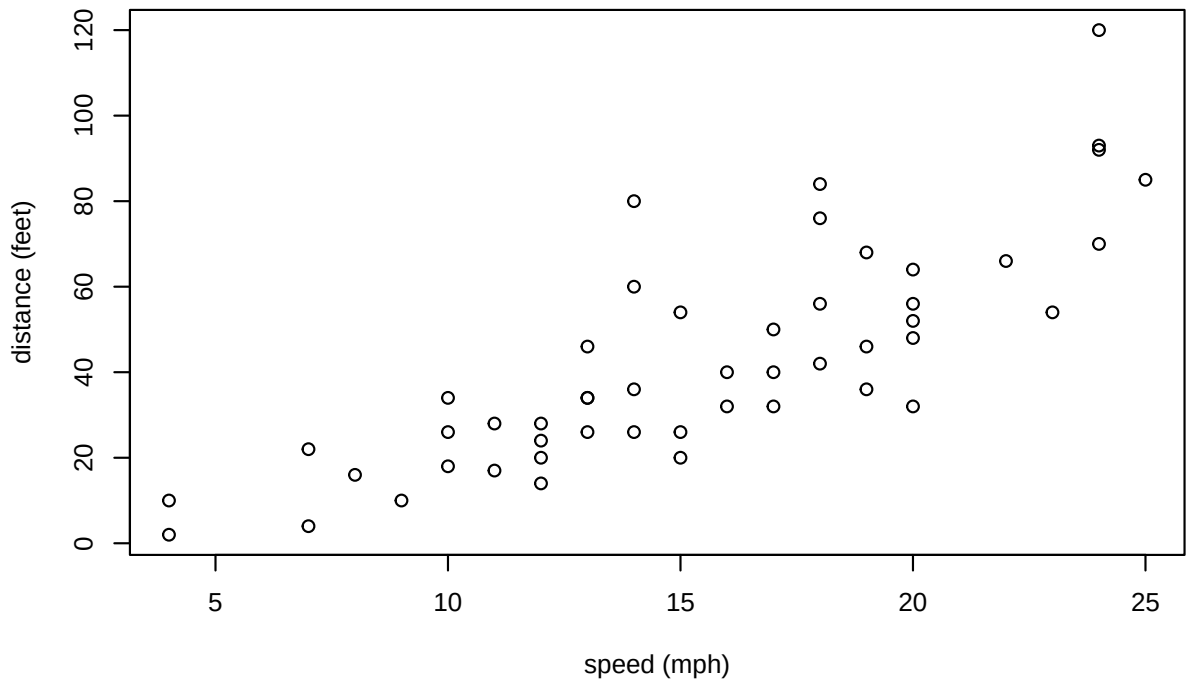

Figure 1: This is my first figure. Figure captions should be located below the figure.

- What are our recommendations for our readers?
- What are worth pursuing further in the future?

Note that we used `enumerate` in Section 1 for a numbered list, but used `itemize` above for a bulleted list.

Remember your project must integrate programming, data management, data analysis, data visualization, and data ethics (collection and use). Make sure these are addressed in this document!

Optional Acknowledgements and Appendix sections are commented out below. Simply remove the % symbol in front of `\section` and the corresponding section will be included. (The $*$ after `\section` prevents these sections from being numbered in the document).

Generally references should appear after acknowledgements, but before any appendices. Be sure to check your bib entries for both formatting and completeness.

Supplemental Materials

Supplemental materials are separate files and document(s), so the purpose of this section is to describe the location and contents of the supplemental material.

All code and data used in this manuscript are available at `my.github.page`.

Descriptions and results of additional supplemental analyses are provided in supplement.pdf, also available at `my.github.page`.

References

A. K. Dwivedi, I. Mallawaarachchi and L. A. Alvarado (2017). "Analysis of small sample size studies using nonparametric bootstrap test with pooled resampling method." *Statistics in Medicine* **36**, 2187–2205.

S. Wild, G. Roglic, A. Green, R. Sicree and H. King (2004). "Global prevalence of diabetes: Estimates for the year 2000 and projections for 2030." *Diabetes Care* **27**, 1047–1053.

Y. Xie (2015). *Dynamic Documents with R and knitr*. Boca Raton, Florida: Chapman and Hall/CRC, 2nd edition. ISBN 978-1498716963.